

Module 2: Dominant Strategies & Strategic Dominance

ECO364-THINKING STRATEGICALLY

What we have learned until now

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- ▶ In Module 1, we studied some of the relevant tools that we have at our disposal to analyze how a decision maker makes choices in the face of uncertainty.
- ▶ We reviewed how to compute probabilities of random events and how to use this to probability to analyze decision problems which involve random events.
- ▶ The problems we analyzed in the last module could be seen as a situation in which a single decision maker was betting against a random outcome.
- ▶ We could think that the aforementioned random outcome was chosen by “nature”.
- ▶ However, when two or more decision makers are interacting with each other in a strategic conflict, the analysis is more complicated.

Motivating Example: Individual Decisions and Games

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A car race is about to start.

Weather can be either sunny or rainy.

The **driver** has to choose his driving style.

Is like if the driver was playing a game against nature, represented by the weather

The probability that is rainy is p .

The probability that is sunny is $1-p$.

How should the driver choose his driving style?

		Weather	
		(p) rainy	($1-p$) sunny
Driver	aggressive	accident	high speed
	conservative	low speed	medium speed

Motivating Example: Individual Decisions and Games

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- ▶ Let's check the elements of this decision problem (as in slide 32 from Module 1)
 - ▶ Decisions: driving style, **aggressive** or **conservative**.
 - ▶ Possible outcomes: **accident**, **high speed**, **medium speed** and **low speed**.
 - ▶ Random events: **rainy** with probability p , **sunny** with probability $1-p$
 - ▶ Preferences: let's say that the driver ranks the outcomes as follows:

High speed > Medium speed > Low speed > accident

	(p) rainy	(1-p) sunny
aggressive	accident	high speed
conservative	low speed	medium speed

Motivating Example: Individual Decisions and Games

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- ▶ High speed>Medium speed>Low speed>accident
- ▶ Given this ranking, we can put numbers (you can think of these numbers as the prize of the race, for example)
 1. High speed~200
 2. Medium speed~150
 3. Low speed~100
 4. Accident~0
- ▶ The table becomes $\Rightarrow$
- ▶ The driver chooses the driving style **aggressive** if and only if its “expected value” is greater than the driving style **conservative**.
- ▶ $0 \cdot p + 200 \cdot (1-p) \geq 100 \cdot p + 150 \cdot (1-p)$

$$1/3 \geq p$$

The driver chooses aggressive if and only if he believes that it will be rainy with less than one third probability!

	(p) rainy	(1-p) sunny
aggressive	0	200
conservative	100	150

Motivating Example: Individual Decisions and Games

- ▶ What if instead of the weather, now the driver is playing against an opponent?
- ▶ The driver has to choose his driving style.
- ▶ The opponent must also choose her driving style.
- ▶ The numbers in the table represent the outcomes, again, think about them like if they were the prizes of the race.
- ▶ How should each player choose his/her action?
- ▶ Notice that the actions of each player, affect the “prize” that the other player can obtain.
- ▶ This is an example of a *strategic conflict*.

		Opponent	
		aggressive	conservative
Driver	aggressive	0, 0	100, 200
	conservative	200, 100	150, 150

Simultaneous Games

DOMINANT STRATEGIES AND STRATEGIC DOMINANCE

A divorce dispute

- ▶ A couple: Alice and Bob decide to divorce. Their total assets are \$500K. They need to reach an agreement of how to divide their assets. Each one of them has two choices: to hire a Brilliant lawyer who charges \$150K for his work, or to hire an Ordinary lawyer who charges a fee of \$50K
- ▶ If both of them hire the same type of lawyer they will split the assets equally: \$250K each. A Brilliant lawyer who faces an Ordinary lawyer will achieve \$375K for her client, leaving \$125K for the other side

The situation can be described by the following table:

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		Bob	
		B	O
Alice	B	100 , 100	225 , 75
	O	75 , 225	200 , 200

We will use this type of table often, so let's analyze its contents in detail

		Bob	
		B	O
Alice	B	100 , 100	225 , 75
	O	75 , 225	200 , 200

Two decision-makers or “players”

Alice and Bob are the ones who are taking the decisions here. We usually use the terminology of “players” to refer to the agents that are making the choices in the situation under consideration.

		Bob		
		B	O	
Alice	B	100 , 100	225 , 75	
	O	75 , 225	200 , 200	

Players' possible decisions (strategies)

Alice and Bob can take two possible decisions each, to hire a Brilliant Lawyer, denoted B, or hire an ordinary lawyer, denoted O. Both players choose their actions simultaneously. We call the choice of action by each player, *a strategy*.

The list of the strategies chosen by each player is called a *strategy profile*. If Alice chooses B and Bob chooses O, the strategy profile is (B,O), where the first element of the list denotes Alice's choice and the second one Bob's.

		Bob	
		B	O
Alice	B	100 , 100	225 , 75
	O	75 , 225	200 , 200

the four
cells in the
table:



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Four possible outcomes (linked to the strategy profiles)

As discussed in the previous slide, a list of strategies for each player is called a strategy profile. Each strategy profile induces an outcome. The outcomes in this case are the four cells in the table.

Considering the story about the split of the assets, the outcomes represent a money amount, but representing outcomes with a “numerical measure can be done in general”

		Bob	
		B	O
Alice	B	100 , 100	225 , 75
	O	75 , 225	200 , 200

the four
cells in the
table:



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Player's payoffs representing preferences over outcomes.

Each outcome induces a payoff, one for each player. In this example, they represent the net amount of money that each player gets after the split and paying their lawyers.

In the description of this situation of the divorce split, it is natural to think about the outcomes in terms of a monetary measure (net amount obtained)

		Bob	
		B	O
Alice	B	100 , 100	225 , 75
	O	75 , 225	200 , 200

the four
cells in the
table:



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In the description of this situation of the divorce split, it is natural to think about the outcomes in terms of a monetary measure (net amount obtained).

However, to do this, we are making some important but subtle assumptions.

Preferences over outcomes can be represented by numbers. In order to do this, each player must be able to rank all outcomes and this ranking should be coherent.

In this example, we are also assuming that each player prefers to have more money to less money.

		Bob	
		B	O
Alice	B	100 , 100	225 , 75
	O	75 , 225	200 , 200

the four
cells in the
table:



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So we have that in this case, each player ranks the four outcomes by looking in which one she or he gets more money.

The preferences for each player over outcomes induces a preference over strategy profiles:

(the first element of the list represents Alice's strategy, the second one Bob's strategy)

Alice: $(B,O) > (O,O) > (B,B) > (O,B)$

Bob: $(O,B) > (O,O) > (B,B) > (B,O)$

Small digression: On representing preferences over outcomes numerically.

- ▶ In this example, the numerical representation of preferences over outcomes is natural: Alice and Bob are splitting a money amount and we are assuming they are interested in how much money they get after the split and after paying their respective lawyers.
- ▶ However, for other strategic situations that we might want to analyze, we must be careful about how we can specify a numerical representation of preferences over outcomes, if at all possible.
- ▶ Furthermore, in this example we are assuming that the **more money the better**, but this **doesn't need to be always the case!**
- ▶ To be able to represent preferences with numbers, we only need that all of the outcomes can be ranked and that this ranking to be consistent.
- ▶ The assumption that “the more the better” depends on the situation we are analyzing.

The situation can be described by the following table:

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		Bob	
		B	O
Alice	B	100 , 100	225 , 75
	O	75 , 225	200 , 200

What will happen?

No matter what Bob decides to do, Alice is always best hiring a Brilliant lawyer:

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If Bob chooses B

Then Alice is better by choosing B, because then the strategy profile (B,B) is associated with a payoff to Alice of 100.

If Alice played O, the strategy profile would be (O,B) and in that case Alice would get 75, which is clearly worse.

		Bob	
		B	O
Alice	B	100 , 100	225 , 75
	O	75 , 225	200 , 200

No matter what Bob decides to do, Alice is always best hiring a Brilliant lawyer:

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If Bob chooses O

		Bob	
		B	O
Alice	B	100 , 100	225 , 75
	O	75 , 225	200 , 200

Then Alice is better by choosing B, because then the strategy profile (B,O) is associated with a payoff to Alice of 225.

If Alice played O, the strategy profile would be (O,O) and in that case Alice would get 200, which is clearly worse.

Similarly, no matter what Alice does, Bob is always better hiring a Brilliant lawyer

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		Bob	
		B	O
Alice	B	100 , 100	225 , 75
	O	75 , 225	200 , 200

If Alice chooses B

Then Bob is better by choosing B, because the strategy profile (B,B) is associated with a payoff to Bob of 100.

If Bob played O instead, we get the profile (B,O), which is associated with a payoff to Bob of 75, and is clearly worse.

Similarly, no matter what Alice does, Bob is always better hiring a Brilliant lawyer

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		Bob	
		B	O
Alice	B	100 , 100	225 , 75
	O	75 , 225	200 , 200

If Alice chooses O

Then Bob is better by choosing B, because the strategy profile (O,B) is associated with a payoff to Bob of 225.

If Bob played O instead, we get the profile (O,O), which is associated with a payoff to Bob of 200, and is clearly worse.

Conclusion...

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- ▶ The only rational outcome is that both Alice and Bob hire Brilliant lawyers and net \$100K each.
- ▶ Who makes the most of it? *The lawyers! Each lawyer gets as a fee 150K.*
- ▶ This might seem paradoxical... Why would this be rational? If they both *agreed* to just hire an Ordinary lawyer, each could go away with 200K...
- ▶ Why this doesn't happen?
- ▶ Consider the way that Alice analyzes the situation. If she believed that Bob would choose a Brilliant lawyer, then she finds it beneficial to also choose a Brilliant lawyer.
- ▶ If she believed that Bob would choose an Ordinary lawyer, again she finds it beneficial to choose a Brilliant lawyer.
- ▶ Similarly for Bob.
- ▶ When we talk about rationality in Game Theory, we mean choosing a course of action that maximizes your payoff.
- ▶ When Alice chooses to hire a Brilliant lawyer, she does that because that strategy maximizes her payoff against all other strategies that Bob could take.
- ▶ Similarly for Bob.
- ▶ In this sense, both Alice and Bob are acting rationally. And this is the paradox: by acting rationally (trying to outsmart the rival), they end up getting less money!

Conclusion...

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- ▶ Remember, to reach the conclusion that both Alice and Bob end up playing the profile (B,B) and each one going away from the divorce with 100K each, we are making the implicit assumption that the *only thing they care about is going away with more money*.
- ▶ It is precisely because of this assumption that both of them will choose to hire Brilliant lawyers.
- ▶ This assumption is represented by the numbers in the payoff table that we presented above.
- ▶ How to represent a situation in which although both Alice and Bob want to go with as much money as possible, they also feel bad if one of them ends up getting more money, i.e. they want a fair division?
- ▶ We would need to alter the payoffs in the table to represent this.
- ▶ Think about payoffs as a mental representation of how much each player values each outcome.
- ▶ You were asked to play **The Grade Game** at the beginning of this module. One of the reasons it didn't have numbers is because I want to see how you would value each outcome. After seeing how the class values each outcome, we will come up with numbers!

Back to Alice and Bob...

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- ▶ To make the point more clear about why is rational to hire a Brilliant lawyer, let's add another strategy for both Alice and Bob.
- ▶ Now, let's assume that both Alice and Bob have the option to negotiate without a lawyer.
- ▶ Assumptions:
 - ▶ When one side is represented by a Brilliant lawyer and the other side is not represented by a lawyer, the lawyer will get for his client \$270K and leave the other side with \$80K
 - ▶ When one side is represented by an Ordinary lawyer and the other side is not represented by a lawyer, the lawyer will get for his client \$265K and leave the other side with \$185K
- ▶ Now the table would be as follows:

Alice \ Bob	Brilliant Lawyer	Average Lawyer	No Lawyer
Brilliant Lawyer	100,100	225,75	270,80
Average Lawyer	75,225	200,200	265,185
No Lawyer	80,270	185,265	250,250

- Let's look at this from the perspective of Alice first:
 - If Alice thinks Bob will choose No Lawyer, she gets more money by hiring a Brilliant lawyer.
 - If Alice thinks Bob will chose an Average lawyer, she gets more money by hiring a Brilliant lawyer.
 - If Alice thinks Bob will choose a Brilliant lawyer, she gets more money by hiring a Brilliant lawyer.
 - Whatever Bob is doing, Alice gets **strictly more money** by hiring a Brilliant lawyer.

Alice \ Bob			
	Brilliant Lawyer	Average Lawyer	No Lawyer
Brilliant Lawyer	100,100	225,75	270,80
Average Lawyer	75,225	200,200	265,185
No Lawyer	80,270	185,265	250,250

- Let's look at this from the perspective of Bob:
 - If Bob thinks Alice will choose No Lawyer, he gets more money by hiring a Brilliant lawyer.
 - If Bob thinks Alice will chose an Average lawyer, he gets more money by hiring a Brilliant lawyer.
 - If Bob thinks Alice will choose a Brilliant lawyer, he gets more money by hiring a Brilliant lawyer.
 - Whatever Alice is doing, Bob gets **strictly more money** by hiring a Brilliant lawyer.

Alice \ Bob			
	Brilliant Lawyer	Average Lawyer	No Lawyer
Brilliant Lawyer	100,100	225,75	270,80
Average Lawyer	75,225	200,200	265,185
No Lawyer	80,270	185,265	250,250

- Definition: A **strictly dominant strategy** for a player is a strategy that is strictly better than any of the other strategies, irrespective of the choice of strategies of the other players.
- By the previous analysis, hiring Brilliant lawyers is a Dominant strategy for both Alice and Bob.
- Even by including the option of Not hiring a lawyer, Hiring a Brilliant lawyer remains a dominant strategy in the expanded game.
- Notice also that the outcome is inferior!

The prisoner's Dilemma

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- ▶ Two suspects in a major crime are held in custody.
- ▶ There is enough evidence too convict each of them of a minor offense, but not enough evidence to convict either of them on a major crime, unless one of them acts as an informer against the other.
- ▶ If they both stay quiet, each will be convicted of the minor offense and spend one year in prison.
- ▶ The DA (district attorney) offers each suspect (individually) a deal:
- ▶ *If the suspect admits, the prosecutor will drop charges for the minor offense.*

- ▶ If one and only one of them admits he will be freed and used as a witness against the other, who
- ▶ will spend life in prison. If both admit each will
- ▶ spend 15 years in prison.

suspect 2 suspect 1		Admit	Quiet
		Admit	Quiet
	Admit	15 years in prison, 15 years in prison	free, Life in prison
	Quiet	Life in prison, free	1 year in prison, 1 year in prison




		suspect 2	
		Admit	Quiet
suspect 1	Admit	15 years in prison, 15 years in prison	free, Life in prison
	Quiet	Life in prison, free	1 year in prison, 1 year in prison

A dominant strategy of each one of them is to admit.
The only Rational Outcome is for them to spend 15 years in prison.

- ▶ We would like to represent preferences with numbers, again, using the fact that larger numbers represent better outcomes.
- ▶ In this case, we will make the natural assumptions that
 - ▶ Both suspects only care about themselves.
 - ▶ They like lower sentences.

suspect 2 suspect 1		Admit	Quiet
		Admit	Quiet
	Admit	-15,-15	0,-100
	Quiet	-100,0	-1,-1

The prisoner's dilemma

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- ▶ Definition: A game is a **Prisoners Dilemma** game if the following two conditions are satisfied:
 1. Every player has a strictly dominant strategy (a strategy that is strictly better than any of the other strategies, irrespective of other players' strategy choices)
 2. When the players choose their dominant strategies the outcome is inferior (namely, there is another outcome that is strictly better for all players)

Game Theory vs Perfect competition (Invisible Hand)

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- ▶ Competition between consumers who choose freely what to buy and producers who choose freely what and how to produce, in both cases in their self interest, will lead to socially efficient outcomes.
- ▶ The invisible hand, referred to by Adam Smith in 1776 (The Wealth of Nations).

“ But man has almost constant occasion for the help of his brethren, and it is in vain for him to expect it from their benevolence only. He will be more likely to prevail if he can interest their self-love in his favour, and shew them that it is for their own advantage to do for him what he requires of them. Whoever offers to another a bargain of any kind, proposes to do this. Give me that which I want, and you shall have this which you want, is the meaning of every such offer; and it is in this manner that we obtain from one another the far greater part of those good offices which we stand in need of. It is not from the benevolence of the butcher, the brewer, or the baker that we expect our dinner, but from their regard to their own interest. We address ourselves, not to their humanity, but to their self-love, and never talk to them of our own necessities, but of their advantages.”

ADAM SMITH, *THE WEALTH OF NATIONS*. CHAPTER 2, P.19

When every producer has only negligible effect on the market, individual self-interest results in social outcomes that cannot be “inferior”.

But most markets are not perfectly competitive: they include “significant” players (firms) whose actions have non-negligible impacts on all other players (imperfect competition). They may result in socially inferior outcomes

Example: The collapse of a cartel

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- ▶ Two countries Iran and Iraq can either produce 2 million or 4 million of barrels of crude oil a day
- ▶ The daily demand for oil is:

$$Q = \frac{120}{P + 5}, \text{ or equivalently } P = \frac{120}{Q} - 5$$

- ▶ P is the market price and Q is the total output
- ▶ Extraction costs are \$2 per barrel in Iran and \$4 per barrel in Iraq

The following tables are for the market price and the profits of the two countries as a result of their outputs:

Table 1: Market Prices

		Iraq	
		$q_2 = 2$	$q_2 = 4$
Iran	$q_1 = 2$	Q=4 P=25	Q=6 P=15
	$q_1 = 4$	Q=6 P=15	Q=8 P=10

Table 2: Profits (Iran, Iraq)



		Iraq	
		$q_2 = 2$	$q_2 = 4$
Iran	$q_1 = 2$	46, 42	26, 44
	$q_1 = 4$	52, 22	32, 24 



The collapse of a cartel.

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- ▶ Clearly, $q_1 = 4$ and $q_2 = 4$ ($Q=8$) are strictly dominant strategies for the corresponding country
- ▶ The outcome of these choices is $(32, 24)$ which is inferior to the “cooperative” outcome $(46, 42)$
- ▶ –i.e., one that could be attained if, for instance, they selected an arbitrator who chose for both countries) and produce 2 million barrels each
- ▶ This cartel game is another example of the Prisoners’ Dilemma

Example:

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Consider the following two persons game:

		2	
		L	R
1	U	4, 4	2, 6
	D	5, 1	x, y

What are the possible values of x and y that turn this game into prisoners' dilemma in which D and R are strictly dominant strategies for players 1 and 2 respectively?

Calculation

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		2	
		L	R
1	U	4, 4	2, 6
	D	5, 1	x, y

- ▶ To make this game a Prisoners' Dilemma we need that D and R become dominant strategies and also that their payoffs x and y respectively are less than 4.
- ▶ D is strictly dominant if $x > 2$.
- ▶ R is strictly dominant if $y > 1$.
- ▶ Thus, we need that

$$2 < x < 4$$

$$1 < y < 4$$

The public transportation Dilemma

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- ▶ Commuters from A to B can choose between public transportation (a bus) and private cars.
- ▶ Denote by p the percentage of commuters by car.
- ▶ Traveling time is:
 - ▶ by bus: $45 + 0.5p$
 - ▶ by car: $30 + 0.5p$

Each driver by its own will contribute a negligible amount to the proportion p by driving his private car, so he will contribute very little to the travel time but save approx. 15 min.
Hence, traveling by car is a dominant strategy.
- ▶ The car ride is shorter regardless of the percentage of car drivers. Therefore, everyone uses the car rather than take the bus.
- ▶ Hence, the trip takes $30 + 0.5 \cdot 100 = 80$ minutes.
- ▶ If all took a bus, it would take $45 + 0.5 \cdot 0 = 45$ minutes.

Is the *More the Better*?

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Increasing (only) the payoffs of one player. Does it make him happier? Not necessarily!

Consider the following simultaneous two person-game G:

- The strategy U is a strictly dominant for Player 1- therefore, he will choose it
- Player 2 knows this and her best reply to U is to play R
- The outcome is (7, 7)

		2	
		L	R
1	U	4, 6	7, 7
	D	3, 5	6, 4

Remark: This game is not a prisoner's dilemma game. Player 1 has a dominant strategy, U, but player 2 has no dominant strategy: if 1 plays U, she should play R, but if 1 plays D, she should play L. However, since U is strictly dominant for 1, player 2 knows this, so is in her best interest to play R.

- ▶ Suppose next that only player 1 is getting a bonus
- ▶ If he plays U his payoff will increase by 1 no matter what 2 chooses
- ▶ If I plays D his payoff will increase by 3, again no matter what 2 chooses
- ▶ What can 1 lose? → All his payoffs increased
- ▶ Is that so? The new game G_1 is:

Game G

		2	
		L	R
1	U	4, 6	7, 7★
	D	3, 5	6, 4

Game G_1

		2	
		L	R
1	U	5, 6	8, 7
	D	6, 5★	9, 4

- Now D is a strictly dominant strategy of 1 and 2 knows this hence 2 will choose L to obtain 5 (and not 4)
- The outcome is (6,5)
- The exclusive bonus of 1 hurts him!

Remark: This last game is not a prisoners' dilemma game. Yes, the outcome (6, 5) is interior to (8,7). Yes, player 1 has a strictly dominant strategy- D. But 2 does not have a dominant strategy.

Adding a dominant strategy to one player

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Does it make him happier? Not necessarily!

- Consider again the game G:

		2	
		L	R
1	U	4, 6	7, 7★
	D	3, 5	6, 4

The only sensible outcome of G_2 is (5, 5) which is inferior to (7, 7)

- Consider next the game G_2 where N is a new strategy of 1 which strictly dominates the other two strategies U and D:

		2	
		L	R
1	U	4, 6	7, 7
	D	3, 5	6, 4
	N	5, 5★	100, 4

More on: Is the More the Better?

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Example: Voting in the Board of Directors

- ▶ A firm has three board members: B1, B2 & B3
- ▶ One of them, B1, is the CEO
- ▶ The board has to choose a policy for the firm out of the following three policies: S1, S2 & S3
- ▶ The policy is selected by a vote
- ▶ Each board member votes a policy in a sealed envelope
- ▶ The policy that obtains a majority wins
- ▶ If no policy has a majority (all three board members select different policies) the policy voted by B1 (the CEO) is implemented
- ▶ **The CEO is therefore endowed with extra power.**

- The following table represents the ranking of the three board members over the possible strategies:

	B_1	B_2	B_3
S_1	E	B	G
S_2	B	G	E
S_3	G	E	B

E- Excellent
G- Good
B- Bad

- The three board members act strategically
- How would they vote?
- What strategy will be implemented

Solution

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- ▶ To solve this game, we need a new definition.
- ▶ Definition: we say that a player has a weakly dominant strategy, if this strategy performs at least as well as any other strategy against whatever the opponent is doing, and it performs strictly better for at least one strategy of the opponent.
- ▶ First observe that it is a weakly dominant strategy for the CEO to vote for S_1 (the best for him).
- ▶ If he does so, S_1 will win unless both B_2 and B_3 vote for S_2 or both vote for S_3 .
- ▶ In these two cases the CEO has no impact on the outcome (his vote does not affect the outcome).

	B_1	B_2	B_3
S_1	E	B	G
S_2	B	G	E
S_3	G	E	B

Consequently, B_1 will vote for S_1 .

- ▶ The two other board members understand that B1 will vote for S1, so after recognizing this fact, their strategic game can be described by the following table.

		B ₃		
		S ₁	S ₂	S ₃
B ₂	S ₁	S ₁	S ₁	S ₁
	S ₂	S ₁	S ₂	S ₁
	S ₃	S ₁	S ₁	S ₃

- ▶ The table shows the winning policy given the decisions of B2 and B1.
- ▶ The ranking of B2 and B3 of S1 is (B, G)- see Table 1
- ▶ Their ranking of S2 is (G, E) and their ranking of S3 is (E, B)

- ▶ We can use this table

	B ₁	B ₂	B ₃
	S ₁	S ₂	S ₃
S ₁	E	B	G
S ₂	B	G	E
S ₃	G	E	B

To write the outcome for B2 and B3 as a function of their decisions.

		B ₃		
		S ₁	S ₂	S ₃
B ₂	S ₁	B, G	B, G	B, G
	S ₂	B, G	G, E	B, G
	S ₃	B, G	B, G	E, B

B ₃ \ B ₂	S ₁	S ₂	S ₃
S ₁	B, G	B, G	B, G
S ₂	B, G	G, E	B, G
S ₃	B, G	B, G	E, B

- ▶ We can use this table to analyze what will happen in the game. On the other hand, we can also assign numbers to the outcomes, so that it is easier to see the preferences. We establish E=3, G=2 and B=1. Then:

	S ₁	S ₂	S ₃
S ₁	1,2	1,2	1,2
S ₂	1,2	2,3	1,2
S ₃	1,2	1,2	3,1

- ▶ Looking at Table 3 we see that S₂ is a weakly dominant strategy for B₃ (he can't do better no matter what B₂ does)
- ▶ Hence B₃ votes for S₂.
- ▶ But then the best reply action of B₂ is also to vote for S₂
- ▶ Consequently S₂ is selected, which is the worst strategy of the CEO

Extra power may hurt!

More on: Is the More the Better?

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Example part 2: Sequential voting

- ▶ Suppose that the votes are open and taken in a sequence.
- ▶ First B_1 announces his choice, then B_2 and then B_3 .
- ▶ The announcement is public.
- ▶ Claim: The winning strategy is S_3 .

	B_1	B_2	B_3	
S_1	E	B	G	E- Excellent G- Good B- Bad
S_2	B	G	E	
S_3	G	E	B	

Explanation of the claim

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- ▶ Suppose first that B_1 votes for S_1 .
- ▶ Then clearly B_2 will not vote for her worst option, S_1 . If she votes for S_3 then B_3 who votes after her will vote for either S_1 or S_2 . In both cases the outcome is S_1 (which is B_2 's worst option). Therefore, B_2 will vote for S_2 and then B_3 will also vote for S_2 . The winning strategy will be S_2 , the worst option for B_1 .
- ▶ Consequently, B_1 will not vote for S_1 (and clearly not for S_2).
- ▶ If B_1 votes for S_3 he induces B_2 to vote for S_3 as well, and the winning strategy is S_3 (a good option for B_1 and excellent for B_2).
- ▶ Consequently, B_1 will vote for S_3 .

Now you try!

Exercise

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1. Show that S_3 will be selected if the order in which the board members vote is (B_1, B_3, B_2) .
2. Show that S_2 will be selected if the order of the votes is (B_2, B_1, B_3) or (B_2, B_3, B_1) or (B_3, B_2, B_1) .
3. Show that S_1 will be selected if the order of the votes is (B_3, B_1, B_2) .

Remark

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- We showed that in case the board members vote simultaneously (that is, every board member when casting his/her vote has no idea of the choices of the other two members) it is a (weakly) dominant strategy for B_1 to vote for S_1 .
- Yet, in a sequential voting, where B_1 announces first his choice, then B_2 announces her vote and then B_3 makes his choice, B_1 votes for S_3 and not for S_1 . Is there a contradiction here?
- The answer is NO. The explanation is a little tricky. In the simultaneous case each member has three strategies: to vote for S_1 or for S_2 or for S_3 , and for B_1 the strategy S_1 dominates the other two strategies against any pair of choices of B_2 and B_3 . However, in the sequential voting where the order is (B_1, B_2, B_3) the situation is different and more complex.

Remark (cont)

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- ▶ We will discuss this in more detail, later when we talk about dynamic games.
- ▶ But for starters: a “strategy” is a complete plan of action for the game.
- ▶ Put yourself in the shoes of B_2 . How many different “plans” do you have?
- ▶ If your answer is 3...
 - ▶ Think Again!
- ▶ Your answer should be $3 \times 3 \times 3$
- ▶ A plan (strategy): what to vote if B_1 votes S_1 , what if she votes S_2 , and what if she votes S_3 . And there are 3 possible answers to each of these questions.
- ▶ How many strategies does B_3 have?!
- ▶ B_3 has 19,683 strategies (3^9).

Remark (cont)

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- ▶ Consider B_2 . Her strategy is now a plan of how to vote for any possible announcement of B_1 , who precedes her:
- ▶ One strategy of B_2 is to vote for S_1 irrespective of B_1 's announcement.
- ▶ Another strategy of B_2 is to vote for S_2 irrespective of B_1 's announcement.
- ▶ A third strategy of B_2 is to vote for S_3 irrespective of B_1 's announcement.
- ▶ These three strategies correspond to the three strategies S_1 , S_2 and S_3 of B_2 in the simultaneous case, where B_2 can not contingent her choice on the choice of B_1 , simply since she does not know his choice.
- ▶ Another strategy of B_2 is to mimic the choice of B_1 . But there are many more. For instance, the strategy (S_2, S_3, S_1) which means that B_2 will vote for S_2 if B_1 votes for S_1 , B_2 will vote for S_3 if B_1 votes for S_2 , and B_2 will vote for S_1 if B_1 votes for S_3 .
- ▶ Any such triple defines a strategy of B_2 , and there are $3 \times 3 \times 3 = 27$

Successive Elimination of Dominated Strategies

55

- Consider the following 5x6 two person game

1 \ 2		a	b	c	d	e	f
A	A	8, 12	12, 8	5, 15	10, 10	11, 9	9, 11
	B	7, 13	11, 9	6, 14	9, 11	8, 12	7, 13
	C	8, 20	26, 4	4, 6	8, 19	7, 7	7, 3
	D	19, 14	12, 14	4, 15	9, 10	10, 10	8, 19
	E	10, 10	22, 6	5, 15	20, 0	18, 2	14, 6

		2					
1		a	b	c	d	e	f
	A	8, 12	12, 8	5, 15	10, 10	11, 9	9, 11
	B	7, 13	11, 9	6, 14	9, 11	8, 12	7, 13
	C	8, 20	26, 4	4, 6	8, 19	7, 7	7, 3
	D	19, 14	12, 14	4, 15	9, 10	10, 10	8, 19
	E	10, 10	22, 6	5, 15	20, 0	18, 2	14, 6

- ▶ First note that b is strictly dominated by c and therefore eliminated.
- ▶ Next note that in the new game C is strictly dominated by E and therefore eliminated
- ▶ Next note that in the new game a is strictly dominated by c therefore eliminated.
- ▶ This is the chain of thought of the players: since b is strictly dominated, 2 will not play it. Since 1 knows that 2 is rational, 1 knows that 2 will not play b. But if 2 doesn't play b, then C is dominated by E, therefore 1 will not play it. But if 2 knows that 1 will not play C, then a becomes strictly dominated by c, thus a will not be played.
- ▶ This iterative process can be continued....
- ▶ Each level of elimination requires a deeper level of “strategic thinking”: I won't play a because my rival won't play C because he knows I won't play b.

		2					
1		a	b	c	d	e	f
	A	8, 12	12, 8	5, 15	10, 10	11, 9	9, 11
	B	7, 13	11, 9	6, 14	9, 11	8, 12	7, 13
	C	8, 20	26, 4	4, 6	8, 19	7, 7	7, 3
	D	19, 14	12, 14	4, 15	9, 10	10, 10	8, 19
	E	10, 10	22, 6	5, 15	20, 0	18, 2	14, 6

- ▶ Next notice that D is strictly dominated by A and thus eliminated
- ▶ Next notice that f is strictly dominated by c and thus eliminated
- ▶ Next notice that e is strictly dominated by c and thus eliminated
- ▶ Next notice that d is strictly dominated by c and thus eliminated
- ▶ Next notice that A and E are both strictly dominated by B
- ▶ The unique outcome is (6, 14)- namely 1 should play B and 2 should play c

Successive elimination of dominated strategies

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- Our conclusion above (that players will/should play B,c and so the “outcome” will be (6,14)) involves a level of complexity of a different order than our conclusion in the Prisoner’s dilemma.
- There, a player does not even need to put herself on the other player’s shoes: no matter what player 2 thinks/knows/does, player 1 is better off by playing Admit.
- Not in this last game!

- Unless player 1 trusts player 2’s ability to realize that she should not play b, player 1 should not discard C!
- And unless player 2 trusts player 1 to be sophisticated enough to realize that she (player 2) will never play b, and so he (player 1) should never play C, player 2 should not consider a to be dominated by c!
- Etc.!

		2					
1		a	b	c	d	e	f
	A	8, 12	12, 8	5, 15	10, 10	11, 9	9, 11
	B	7, 13	11, 9	6, 14	9, 11	8, 12	7, 13
	C	8, 20	26, 4	4, 6	8, 19	7, 7	7, 3
	D	19, 14	12, 14	4, 15	9, 10	10, 10	8, 19
	E	10, 10	22, 6	5, 15	20, 0	18, 2	14, 6

Rationality and common knowledge

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- ▶ We will typically assume that players are “**rational**”:
 - Act in their best interest (which may be altruistic!)
 - Are absolutely able to compute and reason
 - Know all the information in the table
- ▶ Moreover, we will assume that it is “**common knowledge**” that they are rational:
 - They know that the rest of players are rational
 - They know that the rest of players know that they know that they are rational
 - And know that the rest of players know that they know that they know...
- ▶ •Ad infinitum
- ▶ We will discuss more about common knowledge later!